

Posterior Collapse and Two-Boundary Reconstruction Under a Constraint-Only Quantum Retrocausal Hypothesis

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Abstract

We formalize a constraint-only retrocausal hypothesis for quantum measurements in the unitary, finite-dimensional setting. Born probabilities are shown to be time-invariant under unitary propagation. Separately, given a filtered probability space encoding partial measurement records, we define the posterior over terminal outcomes and prove normalization at all intermediate times. Along any rational cofinal sequence approaching the terminal time, the posterior converges almost surely to the conditional expectation given the terminal left-limit σ -algebra, and it collapses to the realized outcome under an explicit terminal identifiability (record-completeness) assumption.

1 Setup

Let $\mathcal{H}$ be a finite-dimensional Hilbert space of dimension d . The system is prepared at time t_i in a density operator $\rho_i \succeq 0$ with $\text{Tr}(\rho_i) = 1$ and evolves to time t via a unitary propagator $U(t, t_i)$.

Remark 1 (Propagator identities). *We assume*

$$U(t_3, t_1) = U(t_3, t_2)U(t_2, t_1), \quad U(t, t) = I, \quad U(t_2, t_1)^\dagger = U(t_1, t_2),$$

for all $t_1 \leq t_2 \leq t_3$.

At a final time $t_f > t_i$, a detector implements a POVM $\{E_k\}_{k \in \mathcal{K}}$ with $E_k \succeq 0$ and $\sum_{k \in \mathcal{K}} E_k = I$, where $\mathcal{K}$ is finite.

Definition 1 (Forward state). *For $t \in [t_i, t_f]$ define*

$$\rho(t) := U(t, t_i)\rho_i U^\dagger(t, t_i).$$

Definition 2 (Backward effect operators). *For each $k \in \mathcal{K}$ and $t \in [t_i, t_f]$ define*

$$\Lambda_k(t) := U^\dagger(t_f, t) E_k U(t_f, t).$$

Remark 2. *Each $\Lambda_k(t) \succeq 0$ and $\sum_{k \in \mathcal{K}} \Lambda_k(t) = I$.*

2 Born Probabilities

Definition 3 (Born distribution). *For $t \in [t_i, t_f]$ define*

$$p_t(k) := \text{Tr}(\rho(t) \Lambda_k(t)).$$

Lemma 1 (Time invariance). *For all $t \in [t_i, t_f]$ and $k \in \mathcal{K}$,*

$$p_t(k) = \text{Tr}(\rho_i U^\dagger(t_f, t_i) E_k U(t_f, t_i)), \quad \sum_{k \in \mathcal{K}} p_t(k) = 1.$$

Proof. Using the propagator identities and cyclicity of trace,

$$\begin{aligned} p_t(k) &= \text{Tr}(U(t, t_i) \rho_i U^\dagger(t, t_i) U^\dagger(t_f, t) E_k U(t_f, t)) \\ &= \text{Tr}(\rho_i U^\dagger(t_f, t_i) E_k U(t_f, t_i)), \end{aligned}$$

which is independent of t . Moreover,

$$\sum_k p_t(k) = \text{Tr}\left(\rho(t) \sum_k \Lambda_k(t)\right) = \text{Tr}(\rho(t)) = 1.$$

□

Definition 4 (Time-invariant Born law). *Define $p(k) := p_t(k)$ for any $t \in [t_i, t_f]$.*

3 Measurement Records and Posterior

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

Remark 3. *Without loss of generality, replace $(\Omega, \mathcal{F}, \mathbb{P})$ by its $\mathbb{P}$ -completion (and keep the same notation).*

Lemma 2 (Sub- σ -algebra completion stays inside a complete ambient space). *Assume $(\Omega, \mathcal{F}, \mathbb{P})$ is complete. If $\mathcal{G} \subseteq \mathcal{F}$ and $\overline{\mathcal{G}}$ denotes the $\mathbb{P}$ -completion of $\mathcal{G}$, then $\overline{\mathcal{G}} \subseteq \mathcal{F}$.*

Proof. If $A \in \overline{\mathcal{G}}$, then there exists $B \in \mathcal{G}$ such that $A \triangle B \subseteq N$ for some $N \in \mathcal{F}$ with $\mathbb{P}(N) = 0$. Completeness implies $A \triangle B \in \mathcal{F}$, hence $A = B \triangle (A \triangle B) \in \mathcal{F}$. □

Let $\{\mathcal{F}_t\}_{t \in \mathbb{Q}, t < t_f}$ be an increasing filtration with

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \quad \text{for all rationals } s \leq t < t_f.$$

For each rational $t < t_f$, let $\overline{\mathcal{F}}_t$ denote the $\mathbb{P}$ -completion of $\mathcal{F}_t$. By Lemma 2, $\overline{\mathcal{F}}_t \subseteq \mathcal{F}$.

Lemma 3 (Completion preserves filtration monotonicity). *If $\mathcal{F}_s \subseteq \mathcal{F}_t$, then $\overline{\mathcal{F}}_s \subseteq \overline{\mathcal{F}}_t$. In particular, $(\overline{\mathcal{F}}_t)_{t \in \mathbb{Q}, t < t_f}$ is an increasing filtration.*

Proof. If $A \in \overline{\mathcal{F}}_s$, then there exists $B \in \mathcal{F}_s$ with $\mathbb{P}(A \triangle B) = 0$. Since $\mathcal{F}_s \subseteq \mathcal{F}_t$, we have $B \in \mathcal{F}_t$, hence $A \in \overline{\mathcal{F}}_t$. $\square$

Definition 5 (Left-limit sigma algebra).

$$\overline{\mathcal{F}}_{t_f-} := \bigvee_{t < t_f, t \in \mathbb{Q}} \overline{\mathcal{F}}_t.$$

Lemma 4 (Completion invariance). *Let $\overline{\mathcal{G}}$ denote the $\mathbb{P}$ -completion of $\mathcal{G} \subseteq \mathcal{F}$. If $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$, then*

$$\mathbb{E}[X \mid \overline{\mathcal{G}}] = \mathbb{E}[X \mid \mathcal{G}] \quad \text{a.s.}$$

Proof. Let $Y := \mathbb{E}[X \mid \mathcal{G}]$. For any $A \in \overline{\mathcal{G}}$ choose $B \in \mathcal{G}$ with $\mathbb{P}(A \triangle B) = 0$. Then

$$\int_A Y \, d\mathbb{P} = \int_B Y \, d\mathbb{P} = \int_B X \, d\mathbb{P} = \int_A X \, d\mathbb{P},$$

so Y satisfies the defining property of $\mathbb{E}[X \mid \overline{\mathcal{G}}]$. $\square$

Definition 6 (Terminal outcome random variable). *Let $K : (\Omega, \mathcal{F}) \rightarrow (\mathcal{K}, 2^{\mathcal{K}})$ be an $\mathcal{F}$ -measurable random variable.*

Axiom 1 (Born-consistent outcome).

$$\mathbb{P}(K = k) = p(k) \quad \text{for all } k \in \mathcal{K}.$$

Definition 7 (Posterior). *For rational $t < t_f$ define*

$$\pi_t(k) := \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_t].$$

Remark 4 (Zero-probability outcomes). *If $p(k) = \mathbb{P}(K = k) = 0$, then $\mathbf{1}_{\{K=k\}} = 0$ a.s., hence $\pi_t(k) = 0$ a.s. for all rational $t < t_f$.*

Definition 8 (Rational cofinal sequence). *A sequence (t_n) is rational cofinal below t_f if $t_n \in \mathbb{Q}$, $t_n < t_f$, $t_n \uparrow t_f$, and for every rational $t < t_f$ there exists n such that $t \leq t_n$.*

Remark 5 (Existence). *Such sequences exist, e.g. choose rationals t_n with $t_f - \frac{1}{n} < t_n < t_f$.*

Lemma 5 (Left-limit join along cofinal sequences). *If (t_n) is rational cofinal below t_f , then*

$$\bigvee_{n \geq 1} \overline{\mathcal{F}}_{t_n} = \bigvee_{t < t_f, t \in \mathbb{Q}} \overline{\mathcal{F}}_t = \overline{\mathcal{F}}_{t_f-}.$$

Proof. Since each t_n is rational, $\bigvee_n \overline{\mathcal{F}}_{t_n} \subseteq \bigvee_{t < t_f, t \in \mathbb{Q}} \overline{\mathcal{F}}_t$. Conversely, fix rational $t < t_f$. By cofinality choose n with $t \leq t_n$; since $(\overline{\mathcal{F}}_t)$ is increasing (Lemma 3), $\overline{\mathcal{F}}_t \subseteq \overline{\mathcal{F}}_{t_n} \subseteq \bigvee_n \overline{\mathcal{F}}_{t_n}$. Taking joins over all such t gives the reverse inclusion. $\square$

Lemma 6 (Lévy upward theorem (discrete-time form)). *Let $(\mathcal{G}_n)_{n \geq 1}$ be an increasing sequence of σ -algebras and let $\mathcal{G} := \bigvee_{n \geq 1} \mathcal{G}_n$. If $X \in L^1$, then*

$$\mathbb{E}[X \mid \mathcal{G}_n] \rightarrow \mathbb{E}[X \mid \mathcal{G}] \quad \text{a.s. and in } L^1.$$

Remark 6. Lemma 6 is a standard martingale convergence result (Lévy's upward theorem); see, e.g., Durrett, Probability: Theory and Examples, Thm. 5.3.2.

Lemma 7 (Posterior normalization). *For all rational $t < t_f$,*

$$0 \leq \pi_t(k) \leq 1, \quad \sum_{k \in \mathcal{K}} \pi_t(k) = 1 \quad \text{a.s.},$$

and for each $k \in \mathcal{K}$, $\{\pi_t(k)\}_{t \in \mathbb{Q}, t < t_f}$ is a martingale: for rationals $s \leq t < t_f$,

$$\mathbb{E}[\pi_t(k) \mid \overline{\mathcal{F}}_s] = \pi_s(k) \quad \text{a.s.}$$

Proof. Since $0 \leq \mathbf{1}_{\{K=k\}} \leq 1$, monotonicity of conditional expectation gives $0 \leq \pi_t(k) \leq 1$ a.s. Linearity yields

$$\sum_{k \in \mathcal{K}} \pi_t(k) = \mathbb{E} \left[\sum_{k \in \mathcal{K}} \mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_t \right] = \mathbb{E}[1 \mid \overline{\mathcal{F}}_t] = 1 \quad \text{a.s.}$$

For $s \leq t$, the tower property gives

$$\mathbb{E}[\pi_t(k) \mid \overline{\mathcal{F}}_s] = \mathbb{E}[\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_t] \mid \overline{\mathcal{F}}_s] = \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_s] = \pi_s(k) \quad \text{a.s.}$$

$\square$

Theorem 1 (Martingale limit). *Let (t_n) be a rational cofinal sequence with $t_n \uparrow t_f$. Then for each $k \in \mathcal{K}$,*

$$\pi_{t_n}(k) \rightarrow \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_{t_f-}] \quad \text{a.s. and in } L^1.$$

Proof. Fix $k \in \mathcal{K}$. Since $\mathbf{1}_{\{K=k\}} \in L^1$, apply Lemma 6 with $\mathcal{G}_n := \overline{\mathcal{F}}_{t_n}$ and $X := \mathbf{1}_{\{K=k\}}$ to obtain

$$\pi_{t_n}(k) = \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_{t_n}] \rightarrow \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \bigvee_n \overline{\mathcal{F}}_{t_n}] \quad \text{a.s. and in } L^1.$$

By Lemma 5, $\bigvee_n \overline{\mathcal{F}}_{t_n} = \overline{\mathcal{F}}_{t_f-}$. $\square$

Axiom 2 (Terminal identifiability). K is $\overline{\mathcal{F}}_{t_f-}$ -measurable.

Corollary 1 (Posterior collapse). For any rational cofinal sequence (t_n) with $t_n \uparrow t_f$,

$$\pi_{t_n}(k) \rightarrow \mathbf{1}_{\{K=k\}} \quad \text{a.s. for all } k \in \mathcal{K}.$$

Proof. Fix k . Since $\mathbf{1}_{\{K=k\}}$ is $\overline{\mathcal{F}}_{t_f-}$ -measurable and integrable,

$$\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_{t_f-}] = \mathbf{1}_{\{K=k\}} \quad \text{a.s.}$$

Combine with Theorem 1. Since $\mathcal{K}$ is finite, intersect over k to obtain the simultaneous statement. $\square$

4 Retrocausal Present State

Definition 9 (Conditioned states). For $t \in [t_i, t_f]$ define

$$\rho_2(t \mid k) = \begin{cases} \frac{\rho^{1/2}(t)\Lambda_k(t)\rho^{1/2}(t)}{p(k)}, & p(k) > 0, \\ I/d, & p(k) = 0. \end{cases}$$

Definition 10 (Random conditioned state). For $t \in [t_i, t_f]$ define the random matrix

$$\rho_2(t \mid K) := \sum_{k \in \mathcal{K}} \mathbf{1}_{\{K=k\}} \rho_2(t \mid k).$$

Lemma 8. For each t and k , $\rho_2(t \mid k) \succeq 0$ and $\text{Tr}(\rho_2(t \mid k)) = 1$.

Proof. If $p(k) = 0$, then $\rho_2(t \mid k) = I/d \succeq 0$ and $\text{Tr}(I/d) = 1$. Assume $p(k) > 0$. Since $\rho(t) \succeq 0$, the positive square root $\rho^{1/2}(t)$ exists and $\rho^{1/2}(t)\Lambda_k(t)\rho^{1/2}(t) \succeq 0$ because $\Lambda_k(t) \succeq 0$. Moreover, by cyclicity of trace,

$$\text{Tr}(\rho^{1/2}(t)\Lambda_k(t)\rho^{1/2}(t)) = \text{Tr}(\rho(t)\Lambda_k(t)) = p_t(k) = p(k),$$

so $\text{Tr}(\rho_2(t \mid k)) = 1$. $\square$

Definition 11 (Retrocausal state). For rational $t < t_f$ define

$$\rho_{\text{RC}}(t) := \sum_{k \in \mathcal{K}} \pi_t(k) \rho_2(t \mid k).$$

Remark 7. We equip $\mathbb{C}^{d \times d} \cong \mathbb{R}^{2d^2}$ with its Borel σ -algebra; equivalently, measurability is entrywise. Since each $\pi_t(k)$ is $\overline{\mathcal{F}}_t$ -measurable and each $\rho_2(t \mid k)$ is deterministic for fixed t, k , $\rho_{\text{RC}}(t)$ is an $\overline{\mathcal{F}}_t$ -measurable $\mathbb{C}^{d \times d}$ -valued random variable.

Lemma 9 (Retrocausal state is a density operator). For each rational $t < t_f$, $\rho_{\text{RC}}(t) \succeq 0$ and $\text{Tr}(\rho_{\text{RC}}(t)) = 1$ a.s.

Proof. By Lemma 8, $\rho_2(t \mid k) \succeq 0$ and $\text{Tr}(\rho_2(t \mid k)) = 1$. Lemma 7 gives $\pi_t(k) \geq 0$ and $\sum_k \pi_t(k) = 1$ a.s. Thus $\rho_{\text{RC}}(t) \succeq 0$ and

$$\text{Tr}(\rho_{\text{RC}}(t)) = \sum_k \pi_t(k) \text{Tr}(\rho_2(t \mid k)) = \sum_k \pi_t(k) = 1 \quad \text{a.s.}$$

□

Corollary 2 (Trace-norm collapse). *For any rational cofinal sequence (t_n) with $t_n \uparrow t_f$,*

$$\|\rho_{\text{RC}}(t_n) - \rho_2(t_n \mid K)\|_1 \rightarrow 0 \quad \text{a.s.}$$

Proof. Write

$$\rho_{\text{RC}}(t_n) - \rho_2(t_n \mid K) = \sum_k (\pi_{t_n}(k) - \mathbf{1}_{\{K=k\}}) \rho_2(t_n \mid k).$$

Since each $\rho_2(t_n \mid k) \succeq 0$ and $\text{Tr}(\rho_2(t_n \mid k)) = 1$, we have $\|\rho_2(t_n \mid k)\|_1 = \text{Tr}(\rho_2(t_n \mid k)) = 1$ (for $A \succeq 0$, $\|A\|_1 = \text{Tr}(A)$), and the triangle inequality yields

$$\|\rho_{\text{RC}}(t_n) - \rho_2(t_n \mid K)\|_1 \leq \sum_k |\pi_{t_n}(k) - \mathbf{1}_{\{K=k\}}|.$$

Define the random variable

$$\pi_{t_n}(K) := \sum_{k \in \mathcal{K}} \pi_{t_n}(k) \mathbf{1}_{\{K=k\}}.$$

On the event $\{K = j\}$, the right-hand side equals

$$(1 - \pi_{t_n}(j)) + \sum_{k \neq j} \pi_{t_n}(k) = 2(1 - \pi_{t_n}(j)),$$

hence pointwise it equals $2(1 - \pi_{t_n}(K))$. By Corollary 1, $\pi_{t_n}(K) \rightarrow 1$ a.s., hence $2(1 - \pi_{t_n}(K)) \rightarrow 0$ a.s., proving the claim. □

5 Conclusion

Born probabilities are time-invariant. Posterior collapse arises purely from Bayesian concentration along rational cofinal sequences, provided the terminal measurement record is complete.